Code No: 124CN **R15**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech II Year II Semester Examinations, July/August - 2021
COMPUTER ORGANIZATION
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Answer any five questions
All questions carry equal marks
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1.a) Discuss three types of CPU organization with examples.
b) Explain 3-address, 2-address, 1-address, zero-address instruction.
X= (A*B)/(C+D-E) [8+7]

2.a) What is meant by program interrupt? Write five examples of external interrupt and five examples of internal interrupt.
b) What is meant by addressing mode? Write about different addressing mode with an example. [7+8]

3.a) Explain the following modes of transfer
i) Programmed I/O ii) Interrupt-initiated I/O
b) Explain asynchronous data transfer. [8+7]

4. What is DMA? Draw the block diagram of DMA controller and explain the use of DMA controller in a computer system. [15]

5.a) What are the difference between static RAM and dynamic RAM in detail?
b) Give a block diagram for an 8M×32 memory using 512k × 8 memory chips. [7+8]

6.a) Explain different mapping mechanism in cache memory.
b) Describe virtual memory address translation and TLB? [8+7]

7.a) Explain the following Addressing Modes with suitable examples
i) Register Indirect ii) Indexed and iii) Register Relative.
b) Explain about various segment registers and their purpose in 8086. [8+7]

8.a) Write an assembly language program to evaluate the sum of first 10 natural numbers.
b) Explain any four flag manipulation instruction in 8086 in detail. [8+7]

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